

Martin's Life work

March 25, 2026

Chapter 1

Help

I need help :(
this is hard!

$$f(x) = mx + b \tag{1.1}$$

but i can fix it!

Chapter 2

Measure Theory

2.1 Measurable Spaces

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2.1.1 Problems

Problem 2.1. *Partition generated by a σ -algebras.*

1. Let $\mathcal{C} = \{A, B, C\}$ be a partition of E . List all the elements of $\sigma\mathcal{C}$
2. let $\mathcal{C}$ be a countable partition of E . show that every element of $\sigma\mathcal{C}$ is a countable union of elements taken from $\mathcal{C}$.
3. Let $E = \mathbb{R}$. Let $\mathcal{C}$ be the collection of all singleton subsets of $\mathbb{R}$, that is, sets that only contain a single point. Show that each element of $\sigma\mathcal{C}$ is either countable, or the complement of a countable set.

Solution.

1. The elements of $\sigma\mathcal{C}$ are defined as the intersection of all sigma-algebras that contain $\mathcal{C}$. Since $\mathcal{C}$ contains 3 subsets of E , as it is a partition, which implies that A, B, C are disjoint and have a union of E . Thus, we can define $\sigma\mathcal{C}$ as the following set:

$$\sigma = \{\emptyset, E, \{A\}, \{B\}, \{C\}, \{A \cap B\}, \{A \cap C\}, \{B \cap C\}\} \quad (2.1)$$

This set contains E , all partitions of closures by E generated from $\mathcal{C}$. Thus, this is the smallest sigma-algebra that contains E generated by $\mathcal{C}$.

2. Since $\mathcal{C}$ is a partition, $\bigcup \mathcal{C} = E$. We need to show that $\sigma\mathcal{C}$ contains only elements from $\mathcal{C}$. We can take ε as the set of all countable unions taken from E . We can show that this is a sigma-algebra by taking its definitions. Since ε contains E the subset of all the partitions, it must also contain the union of none of the partitions, or $\emptyset$. Secondly, by taking all of the unions, we start with the disjoint partitions and then create their unions until we reach E . This will be closed under complement as

by taking countable unions, we will have the sets that are not part of the union as another subset of a different union. Thus, we can argue that this is the same as $\sigma\mathcal{C}$ by the logic of construction. Thus, since each element of $\sigma\mathcal{C}$ is isomorphic to the construction that we obtained from the union of subsets, we have shown that all elements are just a set of unions in $\mathcal{C}$.

3. Define the set $\mathcal{A} = \{A \subset \mathbb{R} : A \text{ is countable or } A^c \text{ is countable}\}$. We want to show that this is a sigma-algebra. Since $\emptyset$ is countable, it is contained within $\mathcal{A}$. $\emptyset \in \mathcal{A}$. The complement of this is the set of reals; $\mathbb{R} = \emptyset^c \in \mathcal{A}$. By definition, if A is countable, then A^c is co-countable. Similarly, if A is co-countable, this implies that A^c is countable. Thus, both sets are contained in $\mathcal{A}$.

Thus, consider a sequence $(A_n)_{n \in \mathbb{N}} \subset \mathcal{A}$. the union of this sequence is $\bigcup_n A_n$. This is split into cases. First, if all A_n are countable, so is the union, which is included in $\mathcal{A}$. If all A_n are co-countable, then consider the complement of the union $(\bigcup_n A_n)^c = \bigcap_n A_n^c$ which is an intersection of countable sets, which is countable. If there is a mix between countable and co-countable sets, they are co-countable, which are contained in $\mathcal{A}$ by the previous definition. Thus, $\bigcup_n A_n \in \mathcal{A}$. Thus, $\mathcal{A}$ is closed under union and complement. Thus, $\mathcal{A}$ is a sigma-algebra. Thus, $\sigma\mathcal{C} \subset \mathcal{A}$.

We want to show that any countable set is a union of the singletons. Since $\mathcal{A}$ contains countable and co-countable sets, $\mathcal{A}$ is a union of singletons, which are by definition contained in $\mathcal{A}$.

$$A = \bigcup_{x \in A} \{x\} \tag{2.2}$$

For co-countable sets in $\mathcal{A}$, we note that the complements are countable, which are contained in the above definition. Since $\mathcal{A}$ consists of countable sets, these are the countable unions of singletons contained in $\sigma\mathcal{C}$. Since $\mathcal{A}$ also consists of co-countable sets, these are the complements of intersections of countable sets in $\sigma\mathcal{C}$. Thus, $\mathcal{A} \subset \sigma\mathcal{C}$.